4	**(a)**	GPE: energy of a <u>mass</u> due to its position in a <u>gravitational field</u>		B1	
		KE: energy (a mass has) due to its motion / speed / velocity		B1	[2]
	(b)	**(i)**	**1.** KE $= \frac{1}{2}mv^2$	C1	
			$= \frac{1}{2} \times 0.4 \times (30)^2$	C1	
			$= 180\,\text{J}$	A1	[3]
			2. $s = 0 + \frac{1}{2} \times 9.81 \times (2.16)^2$ *or* $s = (30\sin 45°)^2 / (2 \times 9.81)$	C1	
			$= 22.88\ (22.9)\,\text{m}$ $\quad = 22.94\ (22.9)\,\text{m}$	A1	[2]
			3. GPE $= mgh$	C1	
			$= 0.4 \times 9.81 \times 22.88 = 89.8\ (90)\ \text{J}$	A1	[2]
		(ii)	**1.** KE = initial KE – GPE = 180 – 90 = 90 J	A1	[1]
			2. (horizontal) velocity is not zero / (object) is still moving / answer explained in terms of conservation of energy	B1	[1]